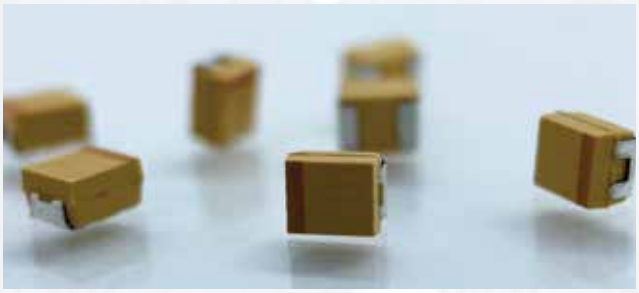
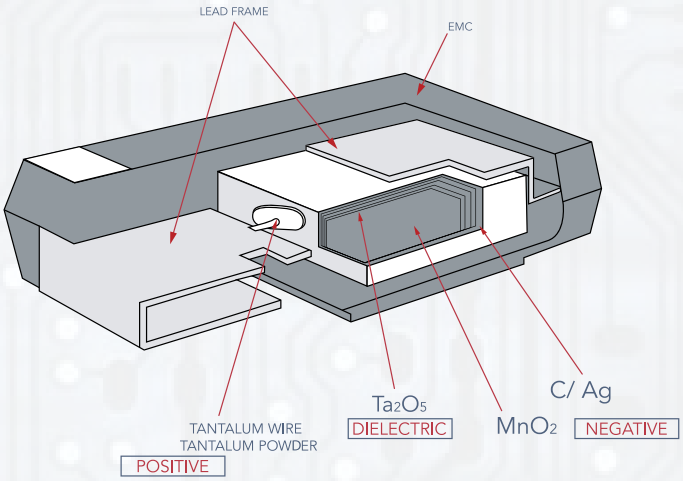
STANDARD TANTALUM CAPACITORS
- TC SERIES -

FEATURES

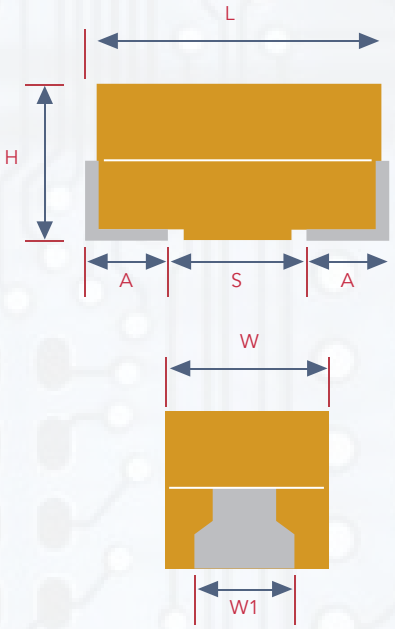
TC Series; The TC Series is designed for hybrid and low profile printed circuit boards where inductance is to be minimized, or where space is at a premium. These units have the further advantage of being compatible with automatic assembly equipment-minus the problems associated with flexible terminal lead wires. Our chip tantalums meet all EIA sizes and are environment friendly, Pb-free.



CONFIGURATION



DIMENSIONS



STANDARD CASE DIMENSIONS

CODE	EIA CODE	EIA METRIC	L±0.20 (0.008)	W+0.20 (0.008) -0.10 (0.004)	H+0.20 (0.008) -0.10 (0.004)	W_1±0.20 (0.008)	A+0.30 (0.012) -0.20 (0.008)	S Min.
A	1206	3216-18	3.20 (0.126)	1.60 (0.063)	1.60 (0.063)	1.20 (0.047)	0.80 (0.031)	1.10 (0.043)
B	1210	3528-21	3.50 (0.138)	2.80 (0.110)	1.90 (0.075)	2.20 (0.087)	0.80 (0.031)	1.40 (0.055)
C	2312	6032-28	6.00 (0.236)	3.20 (0.126)	2.60 (0.102)	2.20 (0.087)	1.30 (0.051)	2.90 (0.114)
D	2917	7343-31	7.30 (0.287)	4.30 (0.169)	2.90 (0.114)	2.40 (0.094)	1.30 (0.051)	4.40 (0.173)
E	2917	7343-43	7.30 (0.287)	4.30 (0.169)	4.10 (0.162)	2.40 (0.094)	1.30 (0.051)	4.40 (0.173)
G	2924	7361-43	7.30 (0.287)	6.10 (0.240)	4.10 (0.162)	3.10 (0.122)	1.30 (0.051)	4.40 (0.173)
H	2924	7361-38	7.30 (0.287)	6.10 (0.240)	3.55 (0.140)	3.10 (0.122)	1.30 (0.051)	4.40 (0.173)

W_1 dimension applies to the termination width for A dimensional area only.

LOW PROFILE CASE DIMENSIONS

CODE	EIA CODE	EIA METRIC	L±0.20 (0.008)	W+0.20 (0.008) -0.10 (0.004)	H MAX.	W_1±0.20 (0.008)	A+0.30 (0.012) -0.20 (0.008)	S Min.
F	2312	6032-20	6.00 (0.236)	3.20 (0.126)	2.00 (0.079)	2.20 (0.087)	1.30 (0.051	2.90 (0.114)
M	1210	3528-15	3.50 (0.138)	2.80 (0.110)	1.50 (0.059)	2.20 (0.087)	0.80 (0.031)	1.40 (0.055)
K	1206	3216-10	3.20 (0.126)	1.60 (0.063)	1.00 (0.039)	1.20 (0.047)	0.80 (0.031)	1.10 (0.043)
P	0805	2012-15	2.05 (0.081)	1.35 (0.053)	1.50 (0.059)	1.00 ±0.10 (0.039±0.004)	0.50 (0.020)	0.85 (0.033)
R	0805	2012-12	2.05 (0.081)	1.30 (0.051)	1.20 (0.047)	1.00 ±0.10 (0.039±0.004)	0.50 (0.020)	0.85 (0.033)
S	1206	3216-12	3.20 (0.126)	1.60 (0.063)	1.20 (0.047)	1.20 (0.047)	0.80 (0.031)	1.10 (0.043)
T	1210	3528-12	3.50 (0.138)	2.80 (0.110)	1.20 (0.047)	2.20 (0.087)	0.80 (0.031)	1.40 (0.055)
U	2312	6032-15	6.00 (0.236)	3.20 (0.126)	1.50 (0.059)	2.20 (0.087)	1.30 (0.051)	2.90 (0.114)
W	2917	7343-15	7.30 (0.287)	4.30 (0.169)	1.50 (0.059)	2.40 (0.094)	1.30 (0.051)	4.40 (0.173)
V	2917	7343-20	7.30 (0.287)	4.30 (0.169)	2.00 (0.079)	2.40 (0.094)	1.30 (0.051)	4.40 (0.173)

W_1 dimension applies to the termination width for A dimensional area only.

PART NUMBERING CODE

TC	K	IA	107	D	T
TYPE OF SERIES	TOLERANCE CODE	RATED VOLTAGE CODE	CAPACITANCE CODE	CASE SIZE CODE	PACKING CODE
	K: 10% M: 20%	2.5 - OB 20 - ID 4 - OG 25 - IE 6.3 - OJ 35 - IV 10 - IA 50 - IX 16 - IC	pF code: 1st two digits represent significant figures 3rd digit represents multiplier (number of zeros to follow)	See case codes above	Pure Tin 7" Reel

Cal-Chip Electronics Inc.
59 Steamwhistle Drive, Ivyland, PA 18974 | p. 215.942.8900 | www.calchip.com | quotes@calchip.com

SPECIFICATIONS

TECHNICAL DATA:	All technical data relate to an ambient temperature of +25°C									
CAPACITANCE RANGE:	0.10 µF to 2200 µF									
CAPACITANCE TOLERANCE:	±10%; ±20%									
RATED VOLTAGE (V_R)	≤ +85°C:	2.5	4	6.3	10	16	20	25	35	50
CATEGORY VOLTAGE (V_C)	≤ +125°C:	1.7	2.7	4	7	10	13	17	23	33
SURGE VOLTAGE (V_S)	≤ +85°C:	3.3	5.2	8	13	20	26	32	46	65
SURGE VOLTAGE (V_S)	≤ +125°C:	2.2	3.4	5	8	13	16	20	28	40
TEMPERATURE RANGE:	-55°C to +125°C									
RELIABILITY:	1% per 1000 hours at 85°C, VR with 0.1Ω/V series impedance, 60% confidence level									
QUALIFICATION:	CECC 30801 - 005 issue 2 EIA 535BAAC for standard case sizes									
TERMINATION FINISHED:	Sn Plating (standard) For AEC-Q200 availability, please contact Cal-Chip Electronics									

STANDARD VALUE AND CASE SIZE

CAP (µF)	R.V.	2.5V	4V (OG)	6.3V (OJ)	10V (IA)	16V (IC)	20V (ID)	25V (IE)	35 (IV)	50V (IX)
0.1	104								A	A
0.15	154								A	A, B
0.22	224								A	A, B
0.33	334								A	A, B
0.47	474							A	A, B	A, B, C
0.68	684							A	A, B	A, B, C
1	105					A	A	A, B	A, B	A, B, C
1.5	155				A	A	A	A, B	A, B, C	B, C, D
2.2	225			A	A	A, B	A, B	A, B	A, B, C	B, C, D
3.3	335			A	A	A, B	A, B	A, B, C	B, C	C, D
4.7	475			A	A, B	A, B	A, B, C	A, B, C	B, C, D	C, D
6.8	685			A, B	A, B	A, B,C	A, B, C	B, C	C, D	C, D
10	106		A	A, B	A, B, C	A, B, C	B, C	B, C, D	C, D, E	D, E, H
15	156		A	A. B	A, B,C	A, B, C	B, C, D	C, D	C, D	D, E, H
22	226		A	A, B, C	A, B, C	A, B, C, D	B, C, D	C, D	D, E	H
33	336	A	A, B	A, B, C	A, B, C, D	B, C, D	C, D	C, D, E	D, E, H	
47	476	A	A, B	A, B, C, D	B, C, D	C, D	C, D, E	D, E	D, E, H	
68	686	A	A, B	B, C, D	B, C, D	C, D	C, D, E	D, E, H	H	
100	107	A,B	A, B, C	B, C, D	B, C, D,E	C, D, E	D, E, H	E, H		
150	157	B	B,C	B, C, D	C, D, E	D,E, H	E, H	H		
220	227	B,D	B, C, D	C, D, E	C, D, E	D, E, H				
330	337	D	C, D	C, D, E	D, E, H	E				
470	477	C,D	C, D, E	D, E, H	E, G, H					
680	687	C,D,E	D, E	D, E, H	E, H					
1000	108	D,E	D, E,H	E, H						
1500	158	D,E,H	E, H							
2200	228	H								

RED DENOTES 20% ONLY

59 Steamwhistle Drive, Ivyland, PA 18974 | p. 215.942.8900 | www.calchip.com | quotes@calchip.com

LOW PROFILE VALUE AND CASE SIZE

CAP (µF)	R.V.	2.5V	4V (OG)	6.3V (OJ)	10V (IA)	16V (IC)	20V (ID)	25V (IE)	35 (IV)	50V (IX)
0.1	104						R, S		R, S	S
0.15	154						R, S	R	R, S	S
0.22	224						R, S	R	R, S	P, R, S
0.33	334						R, S	R	R, S	P, R, S, T
0.47	474						R, S	R, S	R, S, T	S, T
0.68	684					R, S	R, S, T	R, S	P, S, T	
1	105				R, S	R, S, T	R, S, T	P, R, S	P, S, T	U
1.5	155			R, S	R, S	R, S	P, R, S, T	P, S, T	T	U
2.2	225		R, S	R, S	R, S	R, S, T	P, R, S, T	T	T	U
3.3	335		R, S	R, S	K, R, S, T	R, S, T	T	T, U	U	V
4.7	475	R	R, S	R, S, T	R, S, T	K, P, S, T	T	T, U	U	W, V
6.8	685	R	R, S, T	R, S, T	P, R, S, T	S, T	T	U	V	V
10	106	R, S	R, S, T	P, R, S, T	K, P, R, S, T	T, U	U	U	W, V	
15	156	R	R, S, T	K, P, R, S, T	S, T U	T, U	U	V	V	
22	226	P, R	K, P, R, S, T	K, P, S, T, U	T, U	U	U, V	F, V	V	
33	336	K, P, S	K, P, S, T, U	T, U	U	U, V	W, V	F, V		
47	476	P, S	T, U	T, U	M, U, V	U, W, V	W, V	V		
68	686	T	T, U	U	U, V	F, W, V	V			
100	107	T, U	T, U	U, V	U, W, V	F, V				
150	157	T, U	U, V	U, W, V	F, X, V	V				
220	227	U, V	U, W, V	F, W, V	V					
330	337	U, V	F, W, V	V						
470	477	F, V	V	V						
680	687	V	V							
1000	1000	V								

RED DENOTES 20% ONLY



RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
2.5 VOLT RATING @ +85°C (1.7 VOLT RATING @ +125°C												
TC*OB475RT	R	4.7	2.5	85	1.7	125	0.5	6	20	52	47	21
TC*OB685RT	R	6.8	2.5	85	1.7	125	0.5	6	20	52	47	21
TC*OB106RT	R	10	2.5	85	1.7	125	0.5	8	4.5	111	99	44
TC*OB106ST	S	10	2.5	85	1.7	125	0.5	6	8	90	81	36
TC*OB156RT	R	15	2.5	85	1.7	125	0.5	8	4.1	116	104	46
TC*OB226PT	P	22	2.5	85	1.7	125	0.5	8	3.5	131	118	52
TC*OB226RT	R	22	2.5	85	1.7	125	0.5	8	3.8	120	108	48
TC*OB336AT	A	33	2.5	85	1.7	125	0.8	8	1.7	210	189	84
TC*OB336KT	K	33	2.5	85	1.7	125	0.8	8	1.7	196	176	78
TC*OB336PT	P	33	2.5	85	1.7	125	0.7	8	3.5	131	118	52
TC*OB336ST	S	33	2.5	85	1.7	125	0.7	8	1.5	208	187	83
TC*OB476AT	A	47	2.5	85	1.7	125	0.9	6	3	158	142	63
TCMOB476PT	P	47	2.5	85	1.7	125	1.2	12	3.2	137	123	55
TC*OB476ST	S	47	2.5	85	1.7	125	1.2	8	1.6	202	181	81
TC*OB686AT	A	68	2.5	85	1.7	125	1.4	8	1.5	224	201	89
TC*OB686TT	T	68	2.5	85	1.7	125	1.4	8	1.5	231	208	92
TC*OB107AT	A	100	2.5	85	1.7	125	2.5	30	1.4	231	208	93
TC*OB107BT	B	100	2.5	85	1.7	125	2.5	8	1.4	246	222	99
TC*OB107TT	T	100	2.5	85	1.7	125	2.5	15	1.3	248	223	99
TC*OB107UT	U	100	2.5	85	1.7	125	2.5	8	0.4	474	427	190
TC*OB157BT	B	150	2.5	85	1.7	125	3	10	1.6	230	207	92
TCMOB157TT	T	150	2.5	85	1.7	125	3.8	18	1.2	258	232	103
TC*OB157UT	U	150	2.5	85	1.7	125	3.8	8	0.3	548	493	219
TC*OB227BT	B	220	2.5	85	1.7	125	4.4	16	1.6	230	207	92
TC*OB227DT	D	220	2.5	85	1.7	125	5.5	8	0.3	707	636	283
TC*OB227UT	U	220	2.5	85	1.7	125	5.5	8	0.3	548	493	219
TC*OB227VT	V	220	2.5	85	1.7	125	5.5	8	0.3	645	581	258
TC*OB337DT	D	330	2.5	85	1.7	125	8.2	8	0.3	707	636	283
TCMOB337UT	U	330	2.5	85	1.7	125	8.2	12	0.3	548	493	219
TC*OB337VT	V	330	2.5	85	1.7	125	8.2	8	0.3	645	581	258
TC*OB477CT	C	470	2.5	85	1.7	125	9.4	12	0.2	742	667	297
TC*OB477DT	D	470	2.5	85	1.7	125	11.6	8	0.2	866	779	346
TC*OB477FT	F	470	2.5	85	1.7	125	11.8	12	0.3	577	520	231
TC*OB477VT	V	470	2.5	85	1.7	125	11	12	0.2	791	712	316
TC*OB687CT	C	680	2.5	85	1.7	125	17	18	0.2	742	667	297
TC*OB687DT	D	680	2.5	85	1.7	125	17	16	0.2	866	779	346
TC*OB687ET	E	680	2.5	85	1.7	125	17	10	0.2	908	817	363
TC*OB687VT	V	680	2.5	85	1.7	125	17	12	0.2	791	712	316
TCMOB108DT	D	1000	2.5	85	1.7	125	25	20	0.2	866	779	346
TC*OB108ET	E	1000	2.5	85	1.7	125	25	14	0.4	642	578	257
TCMOB108VT	V	1000	2.5	85	1.7	125	25	30	0.2	791	712	316
TC*OB158DT	D	1500	2.5	85	1.7	125	37.5	60	0.2	866	779	346
TC*OB158ET	E	1500	2.5	85	1.7	125	37	20	0.2	908	817	363
TCMOB158HT	H	1500	2.5	85	1.7	125	30	20	0.2	1118	1006	447
TCMOB228HT	H	2200	2.5	85	1.7	125	55	50	0.2	1118	1006	447
4 VOLT RATING @ +85°C (2.7 VOLT RATING @ +125°C												
TC*OG225RT	R	2.2	4	85	2.7	125	0.5	6	25	47	42	19
TC*OG225ST	S	2.2	4	85	2.7	125	0.5	6	25	51	46	20
TC*OG335RT	R	3.3	4	85	2.7	125	0.5	6	20	52	47	21
TC*OG335ST	S	3.3	4	85	2.7	125	0.5	6	18	60	54	24
TC*OG475RT	R	4.7	4	85	2.7	125	0.5	6	12	68	61	27
TC*OG475ST	S	4.7	4	85	2.7	125	0.5	6	10	81	73	32
TC*OG685RT	R	6.8	4	85	2.7	125	0.5	6	5.2	103	93	41
TC*OG685ST	S	6.8	4	85	2.7	125	0.5	6	8	90	81	36
TC*OG685TT	T	6.8	4	85	2.7	125	0.5	6	6	115	104	46
TC*OG106AT	A	10	4	85	2.7	125	0.5	6	6	112	101	45

MS LEVEL 1

* Insert K for +/-10% tolerance and M for +/-20%. | Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts. DCL is measured at a rated voltage after 5 minutes. | Cal-chip reserves the right to substitute a higher voltage rated part Additional values may be available, please contact your sales representative for more information.

RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
4 VOLT RATING @ +85°C (2.7 VOLT RATING @ +125°C												
TC*OG106RT	R	10	4	85	2.7	125	0.5	6	7	89	80	35
TC*OG106ST	S	10	4	85	2.7	125	0.5	6	6	104	94	42
TC*OG106TT	T	10	4	85	2.7	125	0.5	6	5	126	114	51
TC*OG156AT	A	15	4	85	2.7	125	0.6	6	4	137	123	55
TC*OG156RT	R	15	4	85	2.7	125	0.6	8	4	117	106	47
TC*OG156ST	S	15	4	85	2.7	125	0.6	8	4	127	115	51
TC*OG156TT	T	15	4	85	2.7	125	0.6	6	2	200	180	80
TC*OG226AT	A	22	4	85	2.7	125	0.9	6	3.5	146	132	59
TC*OG226KT	K	22	4	85	2.7	125	0.9	8	1.8	190	171	76
TC*OG226PT	P	22	4	85	2.7	125	0.9	8	4	122	110	49
TC*OG226RT	R	22	4	85	2.7	125	0.9	8	3.8	120	108	48
TC*OG226ST	S	22	4	85	2.7	125	0.9	8	3.5	136	123	55
TC*OG226TT	T	22	4	85	2.7	125	0.9	6	1.9	205	185	82
TC*OG336AT	A	33	4	85	2.7	125	1.3	6	3	158	142	63
TC*OG336BT	B	33	4	85	2.7	125	1.3	6	2.8	174	157	70
TC*OG336KT	K	33	4	85	2.7	125	1.3	10	1.7	196	176	78
TCMOG336PT	P	33	4	85	2.7	125	1.3	8	2.8	146	132	59
TC*OG336ST	S	33	4	85	2.7	125	1.3	8	1.7	196	176	78
TC*OG336TT	T	33	4	85	2.7	125	1.3	6	1.7	217	195	87
TC*OG336UT	U	33	4	85	2.7	125	1.3	6	0.6	387	349	155
TC*OG476AT	A	47	4	85	2.7	125	1.9	8	2.6	170	153	68
TC*OG476BT	B	47	4	85	2.7	125	1.9	6	2.4	188	169	75
TC*OG476TT	T	47	4	85	2.7	125	1.9	10	1.6	224	201	89
TC*OG476UT	U	47	4	85	2.7	125	1.9	6	0.5	424	382	170
TC*OG686AT	A	68	4	85	2.7	125	2.7	10	1.5	224	201	89
TC*OG686BT	B	68	4	85	2.7	125	2.7	6	1.8	217	196	87
TC*OG686TT	T	68	4	85	2.7	125	2.7	15	1.5	231	208	92
TC*OG686UT	U	68	4	85	2.7	125	2.7	6	0.4	474	427	190
TC*OG107AT	A	100	4	85	2.7	125	4	30	1.4	231	208	93
TC*OG107BT	B	100	4	85	2.7	125	4	8	0.9	307	277	123
TC*OG107CT	C	100	4	85	2.7	125	4	6	1.3	291	262	116
TCMOG107TT	T	100	4	85	2.7	125	4	14	1.4	239	215	96
TC*OG107UT	U	100	4	85	2.7	125	4	6	0.4	474	427	190
TC*OG157BT	B	150	4	85	2.7	125	6	10	1.5	238	214	95
TC*OG157CT	C	150	4	85	2.7	125	6	6	0.3	606	545	242
TC*OG157UT	U	150	4	85	2.7	125	6	6	0.5	424	382	170
TC*OG157VT	V	150	4	85	2.7	125	6	6	0.4	559	503	224
TC*OG227BT	B	220	4	85	2.7	125	8.8	12	1.1	278	250	111
TC*OG227CT	C	220	4	85	2.7	125	8.8	8	1.2	303	272	121
TC*OG227DT	D	220	4	85	2.7	125	8.8	8	0.9	408	367	163
TC*OG227UT	U	220	4	85	2.7	125	8.8	8	0.3	548	493	219
TC*OG227WT	W	220	4	85	2.7	125	8.8	8	0.3	577	520	231
TC*OG227VT	V	220	4	85	2.7	125	8.8	8	0.3	645	581	258
TC*OG337CT	C	330	4	85	2.7	125	13.2	8	0.3	606	545	242
TC*OG337DT	D	330	4	85	2.7	125	13.2	8	0.9	408	367	163
TC*OG337FT	F	330	4	85	2.7	125	13.2	10	0.3	577	520	231
TC*OG337WT	W	330	4	85	2.7	125	13.2	8	0.3	577	520	231
TC*OG337VT	V	330	4	85	2.7	125	13.2	12	0.4	559	503	224
TC*OG477CT	C	470	4	85	2.7	125	18.8	14	0.3	606	545	242
TC*OG477DT	D	470	4	85	2.7	125	18.8	12	0.9	408	367	163
TC*OG477ET	E	470	4	85	2.7	125	18.8	10	0.5	574	517	230
TC*OG477VT	V	470	4	85	2.7	125	18.8	14	0.4	559	503	224
TC*OG687DT	D	680	4	85	2.7	125	27.2	14	0.5	548	493	219
TC*OG687ET	E	680	4	85	2.7	125	27.2	10	0.9	428	385	171
TCMOG687VT	V	680	4	85	2.7	125	27.2	25	0.2	791	712	316
TC*OG108DT	D	1000	4	85	2.7	125	40	60	0.2	866	779	346
TC*OG108ET	E	1000	4	85	2.7	125	40	14	0.4	642	578	257
TC*OG108HT	H	1000	4	85	2.7	125	40	16	0.2	1118	1006	447
TC*OG158ET	E	1500	4	85	2.7	125	60	30	0.2	908	817	363
TC*OG158HT	H	1500	4	85	2.7	125	60	30	0.2	1118	1006	447

* Insert K for +/-10% tolerance and M for +/-20%. I Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts.
DCL is measured at a rated voltage after 5 minutes. I Cal-chip reserves the right to substitute a higher voltage rated part
Additional values may be available, please contact your sales representative for more information.

Cal-Chip Electronics Inc.
59 Steamwhistle Drive, Ivyland, PA 18974 | p. 215.942.8900 | www.calchip.com | quotes@calchip.com

RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
6.3 VOLT RATING @ +85°C (4 VOLT RATING @125°C)												
TC*OJ155RT	R	1.5	6.3	85	4	125	0.5	6	25	47	42	19
TC*OJ155ST	S	1.5	6.3	85	4	125	0.5	6	25	51	46	20
TC*OJ225AT	A	2.2	6.3	85	4	125	0.5	6	9	91	82	37
TC*OJ225RT	R	2.2	6.3	85	4	125	0.5	6	20	52	47	21
TC*OJ225ST	S	2.2	6.3	85	4	125	0.5	6	18	60	54	24
TC*OJ335AT	A	3.3	6.3	85	4	125	0.5	6	7	104	93	41
TC*OJ335RT	R	3.3	6.3	85	4	125	0.5	6	12	68	61	27
TC*OJ335ST	S	3.3	6.3	85	4	125	0.5	6	9	85	76	34
TC*OJ475AT	A	4.7	6.3	85	4	125	0.5	6	6	112	101	45
TC*OJ475RT	R	4.7	6.3	85	4	125	0.5	6	7	89	80	35
TC*OJ475ST	S	4.7	6.3	85	4	125	0.5	6	7.5	93	84	37
TC*OJ475TT	T	4.7	6.3	85	4	125	0.5	6	6	115	104	46
TC*OJ685AT	A	6.8	6.3	85	4	125	0.5	6	5	122	110	49
TC*OJ685BT	B	6.8	6.3	85	4	125	0.6	6	5	130	117	52
TC*OJ685RT	R	6.8	6.3	85	4	125	0.5	8	7	89	80	35
TC*OJ685ST	S	6.8	6.3	85	4	125	0.5	6	2.6	158	142	63
TC*OJ685TT	T	6.8	6.3	85	4	125	0.5	6	5	126	114	51
TC*OJ106AT	A	10	6.3	85	4	125	0.6	6	4	137	123	55
TC*OJ106BT	B	10	6.3	85	4	125	0.6	6	3	168	151	67
TC*OJ106PT	P	10	6.3	85	4	125	0.6	8	6	100	90	40
TC*OJ106RT	R	10	6.3	85	4	125	0.6	8	6	96	86	38
TC*OJ106ST	S	10	6.3	85	4	125	0.6	8	4	127	115	51
TC*OJ106TT	T	10	6.3	85	4	125	0.6	6	4	141	127	57
TC*OJ156AT	A	15	6.3	85	4	125	0.9	6	3.5	146	132	59
TC*OJ156BT	B	15	6.3	85	4	125	0.9	6	2	206	186	82
TC*OJ156KT	K	15	6.3	85	4	125	0.9	6	2	180	162	72
TC*OJ156PT	P	15	6.3	85	4	125	0.9	8	3.5	131	118	52
TC*OJ156RT	R	15	6.3	85	4	125	0.9	8	4.1	116	104	46
TC*OJ156ST	S	15	6.3	85	4	125	0.9	8	3.5	136	123	55
TC*OJ156TT	T	15	6.3	85	4	125	0.9	6	3.5	151	136	60
TC*OJ226AT	A	22	6.3	85	4	125	1.4	6	3	158	142	63
TC*OJ226BT	B	22	6.3	85	4	125	1.4	6	2.5	184	166	74
TC*OJ226CT	C	22	6.3	85	4	125	1.4	6	2	235	211	94
TC*OJ226KT	K	22	6.3	85	4	125	1.3	10	1.8	190	171	76
TCMOJ226PT	P	22	6.3	85	4	125	1.3	8	3.3	135	121	54
TC*OJ226ST	S	22	6.3	85	4	125	1.3	10	1.8	190	171	76
TC*OJ226TT	T	22	6.3	85	4	125	1.4	8	2.5	179	161	72
TC*OJ226UT	U	22	6.3	85	4	125	1.3	6	0.6	387	349	155
TC*OJ336AT	A	33	6.3	85	4	125	2.1	8	2.2	185	166	74
TC*OJ336BT	B	33	6.3	85	4	125	2.1	6	2.2	197	177	79
TC*OJ336CT	C	33	6.3	85	4	125	2.1	6	1.8	247	222	99
TC*OJ336TT	T	33	6.3	85	4	125	2.1	10	2.5	179	161	72
TC*OJ336UT	U	33	6.3	85	4	125	2	6	0.5	424	382	170
TC*OJ476AT	A	47	6.3	85	4	125	2.8	10	1.6	217	195	87
TC*OJ476BT	B	47	6.3	85	4	125	3	6	2	206	186	82
TC*OJ476CT	C	47	6.3	85	4	125	3	6	1.6	262	236	105
TC*OJ476DT	D	47	6.3	85	4	125	3	6	1.1	369	332	148
TC*OJ476TT	T	47	6.3	85	4	125	2.8	10	1.6	224	201	89
TC*OJ476UT	U	47	6.3	85	4	125	2.8	6	0.5	424	382	170
TC*OJ686BT	B	68	6.3	85	4	125	4	8	0.9	307	277	123
TC*OJ686CT	C	68	6.3	85	4	125	4.3	6	1.5	271	244	108
TC*OJ686DT	D	68	6.3	85	4	125	4.3	6	0.9	408	367	163
TC*OJ686UT	U	68	6.3	85	4	125	4.3	6	1.5	245	220	98
TC*OJ107BT	B	100	6.3	85	4	125	6.3	10	1.7	224	201	89
TC*OJ107CT	C	100	6.3	85	4	125	6.3	6	0.9	350	315	140
TC*OJ107DT	D	100	6.3	85	4	125	6.3	6	0.9	408	367	163
TC*OJ107UT	U	100	6.3	85	4	125	6.3	6	0.9	316	285	126
TC*OJ107VT	V	100	6.3	85	4	125	6.3	6	0.7	423	380	169
TCMOJ157BT	B	150	6.3	85	4	125	9.5	10	1.2	266	240	106
TC*OJ157CT	C	150	6.3	85	4	125	9.5	6	1.3	291	262	116

MS LEVEL 1

* Insert K for +/-10% tolerance and M for +/-20%. | Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts. DCL is measured at a rated voltage after 5 minutes. | Cal-chip reserves the right to substitute a higher voltage rated part Additional values may be available, please contact your sales representative for more information.

RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
6.3 VOLT RATING @ +85°C (4 VOLT RATING @125°C)												
TC*OJ157DT	D	150	6.3	85	4	125	9.5	6	0.9	408	367	163
TC*OJ157UT	U	150	6.3	85	4	125	9	8	0.3	548	493	219
TC*OJ157WT	W	150	6.3	85	4	125	9	6	0.4	500	450	200
TC*OJ157VT	V	150	6.3	85	4	125	9.5	6	0.4	559	503	224
TC*OJ227CT	C	220	6.3	85	4	125	13.9	8	1.2	303	272	121
TC*OJ227DT	D	220	6.3	85	4	125	13.9	8	0.4	612	551	245
TC*OJ227ET	E	220	6.3	85	4	125	13.9	8	0.4	642	578	257
TC*OJ227FT	F	220	6.3	85	4	125	13.2	10	0.3	577	520	231
TC*OJ227WT	W	220	6.3	85	4	125	13.2	8	0.3	577	520	231
TC*OJ227VT	V	220	6.3	85	4	125	13.9	8	0.7	423	380	169
TC*OJ337CT	C	330	6.3	85	4	125	19.8	12	0.5	469	422	188
TC*OJ337DT	D	330	6.3	85	4	125	20.8	8	0.4	612	551	245
TC*OJ337ET	E	330	6.3	85	4	125	20.8	8	0.4	642	578	257
TC*OJ337VT	V	330	6.3	85	4	125	20.8	12	0.4	559	503	224
TC*OJ477DT	D	470	6.3	85	4	125	28	12	0.4	612	551	245
TC*OJ477ET	E	470	6.3	85	4	125	28	10	0.4	642	578	257
TC*OJ477HT	H	470	6.3	85	4	125	28	10	0.4	791	712	316
TC*OJ477VT	V	470	6.3	85	4	125	28.2	20	0.2	791	712	316
TC*OJ687DT	D	680	6.3	85	4	125	40.8	20	0.5	548	493	219
TC*OJ687ET	E	680	6.3	85	4	125	42.8	10	0.5	574	517	230
TC*OJ687HT	H	680	6.3	85	4	125	42.8	10	0.5	707	636	283
TCMOJ108ET	E	1000	6.3	85	4	125	60	20	0.2	908	817	363
TCMOJ108HT	H	1000	6.3	85	4	125	60	16	0.2	1118	1006	447
10 VOLT RATING@+85C(7 VOLT RATING@+125C)												
TC*IA105AT	A	1	10	85	7	125	0.5	4	10	87	78	35
TC*IA105RT	R	1	10	85	7	125	0.5	4	25	47	42	19
TC*IA105ST	S	1	10	85	7	125	0.5	4	25	51	46	20
TC*IA155AT	A	1.5	10	85	7	125	0.5	6	10	87	78	35
TC*IA155RT	R	1.5	10	85	7	125	0.5	6	20	52	47	21
TC*IA155ST	S	1.5	10	85	7	125	0.5	6	20	57	51	23
TC*IA225AT	A	2.2	10	85	7	125	0.5	6	7	104	93	41
TC*IA225RT	R	2.2	10	85	7	125	0.5	6	15	61	54	24
TC*IA225ST	S	2.2	10	85	7	125	0.5	6	12	74	66	29
TC*IA335AT	A	3.3	10	85	7	125	0.5	6	5.5	117	105	47
TC*IA335KT	K	3.3	10	85	7	125	0.5	6	5.5	109	98	43
TC*IA335RT	R	3.3	10	85	7	125	0.5	6	8	83	75	33
TC*IA335ST	S	3.3	10	85	7	125	0.5	6	8	90	81	36
TC*IA335TT	T	3.3	10	85	7	125	0.5	6	6	115	104	46
TC*IA475AT	A	4.7	10	85	7	125	0.5	6	5	122	110	49
TC*IA475BT	B	4.7	10	85	7	125	0.5	6	4	146	131	58
TC*IA475RT	R	4.7	10	85	7	125	0.5	6	9	78	70	31
TC*IA475ST	S	4.7	10	85	7	125	0.5	6	5	114	103	46
TC*IA475TT	T	4.7	10	85	7	125	0.5	6	5	126	114	51
TC*IA685AT	A	6.8	10	85	7	125	0.7	6	4	137	123	55
TC*IA685BT	B	6.8	10	85	7	125	0.7	6	3	168	151	67
TC*IA685PT	P	6.8	10	85	7	125	0.6	6	5	110	99	44
TC*IA685RT	R	6.8	10	85	7	125	0.7	6	5.2	103	93	41
TC*IA685ST	S	6.8	10	85	7	125	0.7	6	4	127	115	51
TC*IA685TT	T	6.8	10	85	7	125	0.7	6	4	141	127	57
TC*IA106AT	A	10	10	85	7	125	1	6	3	158	142	63
TC*IA106BT	B	10	10	85	7	125	1	6	2.1	201	181	80
TC*IA106CT	C	10	10	85	7	125	1	6	2.5	210	189	84
TC*IA106KT	K	10	10	85	7	125	1	6	2.2	172	155	69
TC*IA106PT	P	10	10	85	7	125	1	8	6	100	90	40
TCMIA106RT	R	10	10	85	7	125	1	20	6	96	86	38
TC*IA106ST	S	10	10	85	7	125	1	8	3	147	132	59
TC*IA106TT	T	10	10	85	7	125	1	6	3	163	147	65
TC*IA156AT	A	15	10	85	7	125	1.5	6	3.2	153	138	61
TC*IA156BT	B	15	10	85	7	125	1.5	6	2.8	174	157	70

* Insert K for +/-10% tolerance and M for +/-20%. | Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts. DCL is measured at a rated voltage after 5 minutes. | Cal-chip reserves the right to substitute a higher voltage rated part Additional values may be available, please contact your sales representative for more information.

MS LEVEL 1

RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
10 VOLT RATING@+85C(7 VOLT RATING@+125C)												
TC*IA156CT	C	15	10	85	7	125	1.5	6	2	235	211	94
TC*IA156ST	S	15	10	85	7	125	1.5	6	2	180	162	72
TC*IA156TT	T	15	10	85	7	125	1.5	8	2.8	169	152	68
TC*IA156UT	U	15	10	85	7	125	1.5	6	0.7	359	323	143
TC*IA226AT	A	22	10	85	7	125	2.2	8	3	158	142	63
TC*IA226BT	B	22	10	85	7	125	2.2	6	2.4	188	169	75
TC*IA226CT	C	22	10	85	7	125	2.2	6	1.8	247	222	99
TC*IA226TT	T	22	10	85	7	125	2.2	8	2.2	191	172	76
TC*IA226UT	U	22	10	85	7	125	2.2	6	0.6	387	349	155
TC*IA336AT	A	33	10	85	7	125	3.3	8	1.7	210	189	84
TC*IA336BT	B	33	10	85	7	125	3.3	6	1.8	217	196	87
TC*IA336CT	C	33	10	85	7	125	3.3	6	1.6	262	236	105
TC*IA336DT	D	33	10	85	7	125	3.3	6	1.1	369	332	148
TC*IA336UT	U	33	10	85	7	125	3.3	6	1.6	237	213	95
TC*IA476BT	B	47	10	85	7	125	4.7	8	1	292	262	117
TC*IA476CT	C	47	10	85	7	125	4.7	1.2	1.2	303	272	121
TC*IA476DT	D	47	10	85	7	125	4.7	0.4	0.4	612	551	245
TC*IA476MT	M	47	10	85	7	125	4.7	1	1	283	255	113
TC*IA476UT	U	47	10	85	7	125	4.7	1.4	1.4	254	228	101
TC*IA476VT	V	47	10	85	7	125	4.7	0.5	0.5	500	450	200
TC*IA686BT	B	68	10	85	7	125	6.8	8	1.4	246	222	99
TC*IA686CT	C	68	10	85	7	125	6.8	6	1.3	291	262	116
TC*IA686DT	D	68	10	85	7	125	6.8	6	0.9	408	367	163
TC*IA686UT	U	68	10	85	7	125	6.8	6	1.2	274	246	110
TC*IA686VT	V	68	10	85	7	125	6.8	6	0.9	373	335	149
TC*IA107BT	B	100	10	85	7	125	10	8	1.4	246	222	99
TC*IA107CT	C	100	10	85	7	125	10	8	1.2	303	272	121
TC*IA107DT	D	100	10	85	7	125	10	6	0.9	408	367	163
TC*IA107ET	E	100	10	85	7	125	10	6	0.9	428	385	171
TC*IA107UT	U	100	10	85	7	125	10	6	0.4	474	427	190
TC*IA107WT	W	100	10	85	7	125	10	8	0.9	333	300	133
TC*IA107VT	V	100	10	85	7	125	10	6	0.9	373	335	149
TC*IA157CT	C	150	10	85	7	125	15	8	0.9	350	315	140
TC*IA157DT	D	150	10	85	7	125	15	8	0.9	408	367	163
TC*IA157ET	E	150	10	85	7	125	15	8	0.9	428	385	171
TC*IA157FT	F	150	10	85	7	125	15	10	0.3	577	520	231
TCMIA157WT	W	150	10	85	7	125	15	6	0.3	577	520	231
TC*IA157VT	V	150	10	85	7	125	15	6	1.2	323	290	129
TCKIA227CT	C	220	10	85	7	125	22	16	0.5	469	422	188
TC*IA227DT	D	220	10	85	7	125	22	8	0.5	548	493	219
TC*IA227ET	E	220	10	85	7	125	22	8	0.5	574	517	230
TC*IA227VT	V	220	10	85	7	125	22	10	0.5	500	450	200
TC*IA337DT	D	330	10	85	7	125	33	8	0.9	408	367	163
TC*IA337ET	E	330	10	85	7	125	33	8	0.9	428	385	171
TC*IA337HT	H	330	10	85	7	125	33	10	0.9	527	474	211
TC*IA477ET	E	470	10	85	7	125	47	10	0.5	574	517	230
TC*IA477GT	G	470	10	85	7	125	47	12	0.5	574	517	230
TC*IA477HT	H	470	10	85	7	125	47	10	0.5	707	636	283
TCMIA687ET	E	680	10	85	7	125	68	18	0.4	642	578	257
TCMIA687HT	H	680	10	85	7	125	68	18	0.4	791	712	316
16 VOLT RATING@+85C(10 VOLT RATING@+125C)												
TC*IC684RT	R	0.68	16	85	10	125	0.5	4	25	47	42	19
TC*IC684ST	S	0.68	16	85	10	125	0.5	4	25	51	46	20
TC*IC105AT	A	1	16	85	10	125	0.5	4	11	83	74	33
TC*IC105RT	R	1	16	85	10	125	0.5	4	20	52	47	21
TC*IC105ST	S	1	16	85	10	125	0.5	4	15	66	59	26
TC*IC105TT	T	1	16	85	10	125	0.5	4	5	126	114	51
TC*IC155AT	A	1.5	16	85	10	125	0.5	6	8	97	87	39
TC*IC155RT	R	1.5	16	85	10	125	0.5	6	10	74	67	30
TC*IC155ST	S	1.5	16	85	10	125	0.5	6	12	74	66	29
TC*IC225AT	A	2.2	16	85	10	125	0.5	6	6.5	107	97	43

MS LEVEL 1

* Insert K for +/-10% tolerance and M for +/-20%. | Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts. DCL is measured at a rated voltage after 5 minutes. | Cal-chip reserves the right to substitute a higher voltage rated part Additional values may be available, please contact your sales representative for more information.

Cal-Chip Electronics Inc.
59 Steamwhistle Drive, Ivyland, PA 18974 | p. 215.942.8900 | www.calchip.com | quotes@calchip.com

RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
16 VOLT RATING@+85C (10 VOLT RATING@+125C)												
TC*IC225BT	B	2.2	16	85	10	125	0.5	6	2.3	192	173	77
TC*IC225RT	R	2.2	16	85	10	125	0.5	6	6.5	92	83	37
TC*IC225ST	S	2.2	16	85	10	125	0.5	6	6	104	94	42
TC*IC225TT	T	2.2	16	85	10	125	0.5	6	6.5	111	100	44
TC*IC335AT	A	3.3	16	85	10	125	0.5	6	5	122	110	49
TC*IC335BT	B	3.3	16	85	10	125	0.5	6	4.5	137	124	55
TC*IC335RT	R	3.3	16	85	10	125	0.5	8	5	105	94	42
TC*IC335ST	S	3.3	16	85	10	125	0.5	6	5	114	103	46
TC*IC335TT	T	3.3	16	85	10	125	0.5	6	5	126	114	51
TC*IC475AT	A	4.7	16	85	10	125	0.8	6	4	137	123	55
TC*IC475BT	B	4.7	16	85	10	125	0.8	6	3.5	156	140	62
TC*IC475CT	C	4.7	16	85	10	125	0.8	6	2.4	214	193	86
TC*IC475KT	K	4.7	16	85	10	125	0.8	6	3.1	145	130	58
TC*IC475PT	P	4.7	16	85	10	125	0.8	8	5	110	99	44
TC*IC475ST	S	4.7	16	85	10	125	0.8	8	4	127	115	51
TC*IC475TT	T	4.7	16	85	10	125	0.8	6	3.1	161	145	64
TC*IC685AT	A	6.8	16	85	10	125	1.1	6	3.5	146	132	59
TC*IC685BT	B	6.8	16	85	10	125	1.1	6	2.5	184	166	74
TC*IC685CT	C	6.8	16	85	10	125	1.1	6	2.5	210	189	84
TC*IC685ST	S	6.8	16	85	10	125	1.1	8	2.4	165	148	66
TC*IC685TT	T	6.8	16	85	10	125	1.1	6	3.5	151	136	60
TC*IC106AT	A	10	16	85	10	125	1.6	6	3	158	142	63
TC*IC106BT	B	10	16	85	10	125	1.6	6	2.8	174	157	70
TC*IC106CT	C	10	16	85	10	125	1.6	6	2	235	211	94
TC*IC106DT	D	10	16	85	10	125	1.6	6	1	158	142	63
TC*IC106TT	T	10	16	85	10	125	1.6	8	2.2	191	172	76
TC*IC106UT	U	10	16	85	10	125	1.6	6	2	212	191	85
TC*IC156AT	A	15	16	85	10	125	2.4	6	2	194	174	77
TC*IC156BT	B	15	16	85	10	125	2.4	6	2.5	184	166	74
TC*IC156CT	C	15	16	85	10	125	2.4	6	1.8	247	222	99
TCMIC156TT	T	15	16	85	10	125	2.4	6	2	200	180	80
TC*IC156UT	U	15	16	85	10	125	2.4	6	0.7	359	323	143
TCMIC226AT	A	22	16	85	10	125	3.5	10	2.3	181	163	72
TC*IC226BT	B	22	16	85	10	125	3.5	6	2.3	192	173	77
TC*IC226CT	C	22	16	85	10	125	3.5	6	1	332	298	133
TC*IC226DT	D	22	16	85	10	125	3.5	6	1.1	369	332	148
TC*IC226UT	U	22	16	85	10	125	3.5	6	1.6	237	213	95
TC*IC336BT	B	33	16	85	10	125	5.3	8	2.1	201	181	80
TC*IC336CT	C	33	16	85	10	125	5.3	6	1.5	271	244	108
TC*IC336DT	D	33	16	85	10	125	5.3	6	0.9	408	367	163
TC*IC336UT	U	33	16	85	10	125	5.3	6	1.5	245	220	98
TC*IC336VT	V	33	16	85	10	125	5.3	6	0.9	373	335	149
TC*IC476CT	C	47	16	85	10	125	7.5	6	0.5	469	422	188
TC*IC476DT	D	47	16	85	10	125	7.5	6	0.9	408	367	163
TC*IC476UT	U	47	16	85	10	125	7.5	6	0.4	474	427	190
TC*IC476WT	W	47	16	85	10	125	7.5	6	0.75	365	329	146
TC*IC476VT	V	47	16	85	10	125	7.5	6	0.7	423	380	169
TC*IC686CT	C	68	16	85	10	125	10.9	6	1.3	291	262	116
TC*IC686DT	D	68	16	85	10	125	10.9	6	0.9	408	367	163
TC*IC686FT	F	68	16	85	10	125	10.9	10	0.4	500	450	200
TC*IC686WT	W	68	16	85	10	125	10.9	8	0.6	408	367	163
TC*IC686VT	V	68	16	85	10	125	10.9	6	0.9	373	335	149
TC*IC107CT	C	100	16	85	10	125	16	8	1	332	298	133
TC*IC107DT	D	100	16	85	10	125	16	6	0.6	500	450	200
TC*IC107ET	E	100	16	85	10	125	16	6	0.9	428	385	171
TCMIC107FT	F	100	16	85	10	125	16	10	0.4	500	450	200
TC*IC107VT	V	100	16	85	10	125	16	8	0.9	373	335	149
TC*IC157DT	D	150	16	85	10	125	24	6	0.9	408	367	163
TC*IC157ET	E	150	16	85	10	125	24	8	0.3	742	667	297
TC*IC157HT	H	150	16	85	10	125	24	8	0.5	527	474	211

* Insert K for +/-10% tolerance and M for +/-20%. | Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts. DCL is measured at a rated voltage after 5 minutes. | Cal-chip reserves the right to substitute a higher voltage rated part Additional values may be available, please contact your sales representative for more information.

MS LEVEL 1

RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
16 VOLT RATING@+85C (10 VOLT RATING@+125C)												
TCMIC157VT	V	150	16	85	10	125	24	15	0.3	645	581	258
TCMIC227DT	D	220	16	85	10	125	35.2	10	0.5	548	493	219
TC*IC227ET	E	220	16	85	10	125	35.2	10	0.5	574	517	230
TC*IC227HT	H	220	16	85	10	125	35.2	8	0.9	527	474	211
TCMIC337ET	E	330	16	85	10	125	52.8	30	0.4	642	578	257
20 VOLT RATING@+85C(13 VOLT RATING@+125C)												
TC*ID104RT	R	0.1	20	85	13	125	0.5	4	25	47	42	19
TC*ID104ST	S	0.1	20	85	13	125	0.5	4	25	51	46	20
TC*ID154RT	R	0.15	20	85	13	125	0.5	4	25	47	42	19
TC*ID154ST	S	0.15	20	85	13	125	0.5	4	25	51	46	20
TC*ID224RT	R	0.22	20	85	13	125	0.5	4	25	47	42	19
TC*ID224ST	S	0.22	20	85	13	125	0.5	4	25	51	46	20
TC*ID334RT	R	0.33	20	85	13	125	0.5	4	25	47	42	19
TC*ID334ST	S	0.33	20	85	13	125	0.5	4	25	51	46	20
TC*ID474RT	R	0.47	20	85	13	125	0.5	4	25	47	42	19
TC*ID474ST	S	0.47	20	85	13	125	0.5	4	25	51	46	20
TC*ID684AT	A	0.68	20	85	13	125	0.5	4	12	79	71	32
TC*ID684RT	R	0.68	20	85	13	125	0.5	4	20	52	47	21
TC*ID684ST	S	0.68	20	85	13	125	0.5	4	25	51	46	20
TC*ID684TT	T	0.68	20	85	13	125	0.5	4	15	73	66	29
TC*ID105AT	A	1	20	85	13	125	0.5	4	9	91	82	37
TC*ID105RT	R	1	20	85	13	125	0.5	4	20	52	47	21
TC*ID105ST	S	1	20	85	13	125	0.5	4	12	74	66	29
TC*ID105TT	T	1	20	85	13	125	0.5	4	9	94	85	38
TC*ID155AT	A	1.5	20	85	13	125	0.5	6	6.5	107	97	43
TC*ID155PT	P	1.5	20	85	13	125	0.5	6	9.6	79	71	32
TC*ID155RT	R	1.5	20	85	13	125	0.5	6	9.6	76	68	30
TC*ID155ST	S	1.5	20	85	13	125	0.5	6	5.4	110	99	44
TC*ID155TT	T	1.5	20	85	13	125	0.5	6	6.5	111	100	44
TC*ID225AT	A	2.2	20	85	13	125	0.5	6	5.3	119	107	48
TC*ID225BT	B	2.2	20	85	13	125	0.5	6	3.5	156	140	62
TC*ID225PT	P	2.2	20	85	13	125	0.5	6	8.3	85	77	34
TC*ID225RT	R	2.2	20	85	13	125	0.5	6	6	96	86	38
TC*ID225ST	S	2.2	20	85	13	125	0.5	6	4.5	120	108	48
TC*ID225TT	T	2.2	20	85	13	125	0.5	6	6	115	104	46
TC*ID335AT	A	3.3	20	85	13	125	0.7	6	4.5	129	116	52
TC*ID335BT	B	3.3	20	85	13	125	0.7	6	3	168	151	67
TC*ID335TT	T	3.3	20	85	13	125	0.7	6	3	163	147	65
TC*ID475AT	A	4.7	20	85	13	125	0.9	6	4	137	123	55
TC*ID475BT	B	4.7	20	85	13	125	0.9	6	3	168	151	67
TC*ID475CT	C	4.7	20	85	13	125	0.9	6	2.8	198	178	79
TC*ID475TT	T	4.7	20	85	13	125	0.9	6	3.1	161	145	64
TC*ID685AT	A	6.8	20	85	13	125	1.4	6	2.4	177	159	71
TC*ID685BT	B	6.8	20	85	13	125	1.4	6	2.5	184	166	74
TC*ID685CT	C	6.8	20	85	13	125	1.4	6	2	235	211	94
TC*ID685TT	T	6.8	20	85	13	125	1.4	6	2.6	175	158	70
TC*ID106BT	B	10	20	85	13	125	2	6	2.1	201	181	80
TC*ID106CT	C	10	20	85	13	125	2	6	1.2	303	272	121
TC*ID106DT	D	10	20	85	13	125	2.5	6	1	387	348	155
TC*ID106UT	U	10	20	85	13	125	2	6	1.9	218	196	87
TC*ID156BT	B	15	20	85	13	125	3	6	2	206	186	82
TC*ID156CT	C	15	20	85	13	125	3	6	1.7	254	229	102
TC*ID156DT	D	15	20	85	13	125	3	6	1.1	369	332	148
TC*ID156UT	U	15	20	85	13	125	3	6	1.7	230	207	92
TC*ID226BT	B	22	20	85	13	125	4.4	6	1.8	217	196	87
TC*ID226CT	C	22	20	85	13	125	4.4	6	1.6	262	236	105
TC*ID226DT	D	22	20	85	13	125	4.4	6	0.9	408	367	163
TC*ID226UT	U	22	20	85	13	125	4.4	6	1.6	237	213	95
TC*ID226VT	V	22	20	85	13	125	4.4	6	0.9	373	335	149
TC*ID336CT	C	33	20	85	13	125	6.6	6	1.5	271	244	108
TC*ID336DT	D	33	20	85	13	125	6.6	6	0.9	408	367	163

MS LEVEL 1

* Insert K for +/-10% tolerance and M for +/-20%. | Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts. DCL is measured at a rated voltage after 5 minutes. | Cal-chip reserves the right to substitute a higher voltage rated part Additional values may be available, please contact your sales representative for more information.

Cal-Chip Electronics Inc.
59 Steamwhistle Drive, Ivyland, PA 18974 | p. 215.942.8900 | www.calchip.com | quotes@calchip.com

RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
20 VOLT RATING@+85C(13 VOLT RATING@+125C)												
TC*ID336WT	W	33	20	85	13	125	6.6	6	0.5	447	402	179
TC*ID336VT	V	33	20	85	13	125	6.6	6	0.6	456	411	183
TC*ID476CT	C	47	20	85	13	125	9.4	6	0.5	469	422	188
TC*ID476DT	D	47	20	85	13	125	9.4	6	0.9	408	367	163
TC*ID476ET	E	47	20	85	13	125	9.4	6	0.9	428	385	171
TC*ID476WT	W	47	20	85	13	125	9.4	6	0.4	500	450	200
TC*ID476VT	V	47	20	85	13	125	9.4	6	0.9	373	335	149
TC*ID686CT	C	68	20	85	13	125	13.6	8	0.5	469	422	188
TC*ID686DT	D	68	20	85	13	125	13.6	6	0.4	612	551	245
TC*ID686ET	E	68	20	85	13	125	13.6	6	0.9	428	385	171
TC*ID686VT	V	68	20	85	13	125	13.6	6	0.9	373	335	149
TC*ID107DT	D	100	20	85	13	125	20	6	0.5	548	493	219
TC*ID107ET	E	100	20	85	13	125	20	6	0.4	642	578	257
TC*ID107HT	H	100	20	85	13	125	20	8	0.9	527	474	211
TC*ID157ET	E	150	20	85	13	125	30	8	0.3	742	667	297
TC*ID157HT	H	150	20	85	13	125	30	8	0.3	913	822	365
25 VOLT RATING@+85C(17 VOLT RATING@+125C)												
TC*IE154RT	R	0.15	25	85	17	125	0.5	4	24	48	43	19
TC*IE224RT	R	0.22	25	85	17	125	0.5	4	21	51	46	20
TC*IE334AT	A	0.33	25	85	17	125	0.5	4	17	71	64	28
TC*IE334RT	R	0.33	25	85	17	125	0.5	4	17	57	51	23
TC*IE474AT	A	0.47	25	85	17	125	0.5	4	14	73	66	29
TC*IE474RT	R	0.47	25	85	17	125	0.5	4	15	61	54	24
TC*IE474ST	S	0.47	25	85	17	125	0.5	4	9	85	76	34
TC*IE684AT	A	0.68	25	85	17	125	0.5	4	10	87	78	35
TC*IE684RT	R	0.68	25	85	17	125	0.5	4	13	65	59	26
TC*IE684ST	S	0.68	25	85	17	125	0.5	4	8	90	81	36
TC*IE105AT	A	1	25	85	17	125	0.5	4	8	97	87	39
TC*IE105BT	B	1	25	85	17	125	0.5	4	7.5	114	103	46
TC*IE105PT	P	1	25	85	17	125	0.5	4	11	74	66	30
TC*IE105RT	R	1	25	85	17	125	0.5	4	8	83	75	33
TC*IE105ST	S	1	25	85	17	125	0.5	4	8	90	81	36
TC*IE155AT	A	1.5	25	85	17	125	0.5	6	7.5	100	90	40
TC*IE155BT	B	1.5	25	85	17	125	0.5	6	5	130	117	52
TC*IE155PT	P	1.5	25	85	17	125	0.5	6	9.6	79	71	32
TC*IE155ST	S	1.5	25	85	17	125	0.5	6	5.4	110	99	44
TC*IE155TT	T	1.5	25	85	17	125	0.5	6	5	126	114	51
TC*IE225AT	A	2.2	25	85	17	125	0.6	6	7	104	93	41
TC*IE225BT	B	2.2	25	85	17	125	0.6	6	4.5	137	124	55
TC*IE225TT	T	2.2	25	85	17	125	0.6	6	4.5	133	120	53
TC*IE335AT	A	3.3	25	85	17	125	0.8	6	3.7	142	128	57
TC*IE335BT	B	3.3	25	85	17	125	0.8	6	3.5	156	140	62
TC*IE335CT	C	3.3	25	85	17	125	0.8	6	2.8	198	178	79
TC*IE335TT	T	3.3	25	85	17	125	0.8	6	3.5	151	136	60
TC*IE335UT	U	3.3	25	85	17	125	0.8	6	1.6	237	213	95
TC*IE475AT	A	4.7	25	85	17	125	1.2	6	3.1	156	140	62
TC*IE475BT	B	4.7	25	85	17	125	1.2	6	1.5	238	214	95
TC*IE475CT	C	4.7	25	85	17	125	1.2	6	2.4	214	193	86
TC*IE475TT	T	4.7	25	85	17	125	1.2	6	3.1	161	145	64
TC*IE475UT	U	4.7	25	85	17	125	1.2	6	1.2	274	246	110
TC*IE685BT	B	6.8	25	85	17	125	1.7	6	2.8	174	157	70
TC*IE685CT	C	6.8	25	85	17	125	1.7	6	2	235	211	94
TC*IE685UT	U	6.8	25	85	17	125	1.7	6	2	212	191	85
TC*IE106BT	B	10	25	85	17	125	2.5	6	2.5	184	166	74
TC*IE106CT	C	10	25	85	17	125	2.5	6	1.8	247	222	99
TC*IE106DT	D	10	25	85	17	125	2.5	6	1.2	354	318	141
TC*IE106UT	U	10	25	85	17	125	2.5	6	1.8	224	201	89
TC*IE156CT	C	15	25	85	17	125	3.8	6	1.6	262	236	105
TC*IE156DT	D	15	25	85	17	125	3.8	6	1	387	349	155
TC*IE156VT	V	15	25	85	17	125	3.8	6	1	354	318	141
TC*IE226CT	C	22	25	85	17	125	5.5	6	1.4	280	252	112
TC*IE226DT	D	22	25	85	17	125	5.5	6	0.9	408	367	163

* Insert K for +/-10% tolerance and M for +/-20%. | Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts. DCL is measured at a rated voltage after 5 minutes. | Cal-chip reserves the right to substitute a higher voltage rated part Additional values may be available, please contact your sales representative for more information.

RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
25 VOLT RATING@+85C(17 VOLT RATING@+125C)												
TC*IE226FT	F	22	25	85	17	125	5.5	6	1	316	285	126
TC*IE226VT	V	22	25	85	17	125	5.5	6	0.8	395	356	158
TC*IE336CT	C	33	25	85	17	125	8.3	6	0.9	350	315	140
TC*IE336DT	D	33	25	85	17	125	8.3	6	0.9	408	367	163
TC*IE336ET	E	33	25	85	17	125	8.3	6	0.9	428	385	171
TC*IE336FT	F	33	25	85	17	125	8.3	6	0.9	333	300	133
TC*IE336VT	V	33	25	85	17	125	8.3	6	0.5	500	450	200
TC*IE476DT	D	47	25	85	17	125	11.8	6	0.9	408	367	163
TC*IE476ET	E	47	25	85	17	125	11.8	6	0.9	428	385	171
TC*IE476VT	V	47	25	85	17	125	11.8	6	0.9	373	335	149
TC*IE686DT	D	68	25	85	17	125	17	6	0.9	408	367	163
TC*IE686ET	E	68	25	85	17	125	17	6	0.9	428	385	171
TC*IE686HT	H	68	25	85	17	125	17	6	0.9	527	474	211
TC*IE107ET	E	100	25	85	17	125	25	10	0.3	742	667	297
TC*IE107HT	H	100	25	85	17	125	25	8	0.4	791	712	316
TCMIE157HT	H	150	25	85	17	125	37.5	10	0.4	791	712	316
35 VOLT RATING@+85C(23 VOLT RATING@+125C)												
TC*IV104AT	A	0.1	35	85	23	125	0.5	4	24	56	50	22
TC*IV104RT	R	0.1	35	85	23	125	0.5	4	29	44	39	17
TC*IV104ST	S	0.1	35	85	23	125	0.5	4	24	52	47	21
TC*IV154AT	A	0.15	35	85	23	125	0.5	4	21	60	54	24
TC*IV154RT	R	0.15	35	85	23	125	0.5	4	24	48	43	19
TC*IV154ST	S	0.15	35	85	23	125	0.5	4	21	56	50	22
TC*IV224AT	A	0.22	35	85	23	125	0.5	4	18	65	58	26
TC*IV224RT	R	0.22	35	85	23	125	0.5	4	21	51	46	20
TC*IV224ST	S	0.22	35	85	23	125	0.5	4	18	60	54	24
TC*IV334AT	A	0.33	35	85	23	125	0.5	4	15	71	64	28
TC*IV334RT	R	0.33	35	85	23	125	0.5	4	17	57	51	23
TC*IV334ST	S	0.33	35	85	23	125	0.5	4	15	66	59	26
TC*IV474AT	A	0.47	35	85	23	125	0.5	4	12	79	71	32
TC*IV474BT	B	0.47	35	85	23	125	0.5	4	10	92	83	37
TC*IV474RT	R	0.47	35	85	23	125	0.5	4	15	61	54	24
TC*IV474ST	S	0.47	35	85	23	125	0.5	4	12	74	66	29
TC*IV474TT	T	0.47	35	85	23	125	0.5	4	10	89	80	36
TC*IV684AT	A	0.68	35	85	23	125	0.5	4	8	97	87	39
TC*IV684BT	B	0.68	35	85	23	125	0.5	4	8	103	93	41
TC*IV684PT	P	0.68	35	85	23	125	0.5	4	13	68	61	27
TC*IV684ST	S	0.68	35	85	23	125	0.5	4	8	90	81	36
TC*IV684TT	T	0.68	35	85	23	125	0.5	4	8	100	90	40
TC*IV105AT	A	1	35	85	23	125	0.5	4	7.5	100	90	40
TC*IV105BT	B	1	35	85	23	125	0.5	4	6.5	114	103	46
TC*IV105PT	P	1	35	85	23	125	0.5	4	11	74	66	30
TC*IV105ST	S	1	35	85	23	125	0.5	4	7.5	93	84	37
TC*IV105TT	T	1	35	85	23	125	0.5	4	6.5	111	100	44
TC*IV155AT	A	1.5	35	85	23	125	0.5	6	7.5	100	90	40
TC*IV155BT	B	1.5	35	85	23	125	0.5	6	5.2	128	115	51
TC*IV155CT	C	1.5	35	85	23	125	0.5	6	4.5	156	141	63
TC*IV155TT	T	1.5	35	85	23	125	0.5	6	5.2	124	112	50
TC*IV225AT	A	2.2	35	85	23	125	0.8	6	4.5	129	116	52
TC*IV225BT	B	2.2	35	85	23	125	0.8	6	4.2	142	128	57
TC*IV225CT	C	2.2	35	85	23	125	0.8	6	3.5	177	160	71
TC*IV225TT	T	2.2	35	85	23	125	0.8	6	4.2	138	124	55
TC*IV335BT	B	3.3	35	85	23	125	1.2	6	3.5	156	140	62
TC*IV335CT	C	3.3	35	85	23	125	1.2	6	2.5	210	189	84
TC*IV335UT	U	3.3	35	85	23	125	1.2	6	1.6	237	213	95
TC*IV475BT	B	4.7	35	85	23	125	1.6	6	3.1	166	149	66
TC*IV475CT	C	4.7	35	85	23	125	1.6	6	2.2	224	201	89
TC*IV475DT	D	4.7	35	85	23	125	1.6	6	1.5	316	285	126
TC*IV475UT	U	4.7	35	85	23	125	1.6	6	2.2	202	182	81
TC*IV685CT	C	6.8	35	85	23	125	2.4	6	1.8	247	222	99
TC*IV685DT	D	6.8	35	85	23	125	2.4	6	1.3	340	306	136
TC*IV685VT	V	6.8	35	85	23	125	2.3	6	0.9	373	335	149

MS LEVEL 1

* Insert K for +/-10% tolerance and M for +/-20%. | Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts. DCL is measured at a rated voltage after 5 minutes. | Cal-chip reserves the right to substitute a higher voltage rated part Additional values may be available, please contact your sales representative for more information.

Cal-Chip Electronics Inc.
59 Steamwhistle Drive, Ivyland, PA 18974 | p. 215.942.8900 | www.calchip.com | quotes@calchip.com

RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
35 VOLT RATING@+85C(23 VOLT RATING@+125C)												
TC*IV106CT	C	10	35	85	23	125	3.5	6	1.6	262	236	105
TC*IV106DT	D	10	35	85	23	125	3.5	6	1	387	349	155
TC*IV106ET	E	10	35	85	23	125	3.5	6	0.9	428	385	171
TC*IV106WT	W	10	35	85	23	125	3.5	6	0.7	378	340	151
TC*IV106VT	V	10	35	85	23	125	3.5	6	1	354	318	141
TC*IV156CT	C	15	35	85	23	125	5.3	6	1.4	280	252	112
TC*IV156DT	D	15	35	85	23	125	5.3	6	0.9	408	367	163
TC*IV156VT	V	15	35	85	23	125	5.3	6	0.6	456	411	183
TC*IV226DT	D	22	35	85	23	125	7.7	6	0.9	408	367	163
TC*IV226ET	E	22	35	85	23	125	7.7	6	0.5	574	517	230
TC*IV226VT	V	22	35	85	23	125	7.7	6	0.5	500	450	200
TC*IV336DT	D	33	35	85	23	125	11.6	6	0.9	408	367	163
TC*IV336ET	E	33	35	85	23	125	11.6	6	0.9	428	385	171
TC*IV336HT	H	33	35	85	23	125	11.6	6	0.5	707	636	283
TC*IV476DT[1]	D	47	35	85	23	125	16.5	6	0.9	408	367	163
TC*IV476ET	E	47	35	85	23	125	16.5	6	0.9	428	385	171
TC*IV476HT	H	47	35	85	23	125	16.5	6	0.4	791	712	316
TC*IV686HT	H	68	35	85	23	125	23.8	6	0.5	707	636	283
50 VOLT RATING@+85C(33 VOLT RATING@+125C)												
TC*IX104AT	A	0.1	50	85	33	125	0.5	4	22	58	53	23
TC*IX104ST	S	0.1	50	85	33	125	0.5	4	19	58	53	23
TC*IX154AT	A	0.15	50	85	33	125	0.5	4	15	71	64	28
TC*IX154BT	B	0.15	50	85	33	125	0.5	4	17	71	64	28
TC*IX154ST	S	0.15	50	85	33	125	0.5	4	16	64	57	25
TC*IX224AT	A	0.24	50	85	33	125	0.5	4	18	65	58	26
TC*IX224BT	B	0.24	50	85	33	125	0.5	4	14	78	70	31
TC*IX224PT	P	0.24	50	85	33	125	0.5	4	17	59	53	24
TC*IX224RT	R	0.24	50	85	33	125	0.5	4	17	57	51	23
TC*IX224ST	S	0.24	50	85	33	125	0.5	4	13	71	64	28
TC*IX334AT	A	0.33	50	85	33	125	0.5	4	17	66	60	27
TC*IX334BT	B	0.33	50	85	33	125	0.5	4	12	84	76	34
TC*IX334PT	P	0.33	50	85	33	125	0.5	4	17	59	53	24
TCMIX334RT	R	0.33	50	85	33	125	0.5	4	17	57	51	23
TC*IX334ST	S	0.33	50	85	33	125	0.5	4	11	77	69	31
TC*IX334TT	T	0.33	50	85	33	125	0.5	4	11	85	77	34
TC*IX474AT	A	0.47	50	85	33	125	0.5	4	9.5	89	80	36
TC*IX474BT	B	0.47	50	85	33	125	0.5	4	9.5	95	85	38
TC*IX474CT	C	0.47	50	85	33	125	0.5	4	8	117	106	47
TC*IX474ST	S	0.47	50	85	33	125	0.5	4	9.5	83	74	33
TC*IX474TT	T	0.47	50	85	33	125	0.5	4	9.5	92	83	37
TC*IX684AT	A	0.68	50	85	33	125	0.5	4	7.9	97	88	39
TC*IX684BT	B	0.68	50	85	33	125	0.5	4	8	103	93	41
TC*IX684CT	C	0.68	50	85	33	125	0.5	4	7	125	113	50
TC*IX105AT	A	1	50	85	33	125	0.5	4	6.6	107	96	43
TC*IX105BT	B	1	50	85	33	125	0.5	6	7	110	99	44
TC*IX105CT	C	1	50	85	33	125	0.5	4	5.5	141	127	57
TC*IX105UT	U	1	50	85	33	125	0.5	6	4.4	143	129	57
TC*IX155BT	B	1.5	50	85	33	125	0.8	8	5.4	125	113	50
TC*IX155CT	C	1.5	50	85	33	125	0.8	6	4.5	156	141	63
TC*IX155DT	D	1.5	50	85	33	125	0.8	6	4	194	174	77
TC*IX155UT	U	1.5	50	85	33	125	0.8	6	3.1	170	153	68
TC*IX225BT	B	2.5	50	85	33	125	1.1	8	4.5	137	124	55
TC*IX225CT	C	2.5	50	85	33	125	1.1	8	2.5	210	189	84
TC*IX225DT	D	2.5	50	85	33	125	1.1	6	2.5	245	220	98
TC*IX225UT	U	2.5	50	85	33	125	1.1	8	2.5	190	171	76
TC*IX335CT	C	3.3	50	85	33	125	1.6	6	2.5	210	189	84
TC*IX335DT	D	3.3	50	85	33	125	1.7	6	2	274	246	110
TC*IX335VT	V	3.3	50	85	33	125	1.7	4	1.5	289	260	115
TC*IX475CT	C	4.7	50	85	33	125	2.4	6	1.4	280	252	112

[1] MSL Level 3 for both K & M Tolerance

* Insert K for +/-10% tolerance and M for +/-20%. | Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts. DCL is measured at a rated voltage after 5 minutes. | Cal-chip reserves the right to substitute a higher voltage rated part Additional values may be available, please contact your sales representative for more information.

MS LEVEL 1

RATINGS & PART NUMBER REFERENCE

PART NUMBER	CASE SIZE	CAPACITANCE (µF)	RATED VOLTAGE (V)	RATED TEMP. (C)	CATEGORY VOLTAGE (V)	CATEGORY TEMP. (°C)	DC LEAKAGE (UA) @+25°C MAX	DF (%) @+25°C 120Hz MAX	ESR (Ohm) @+25°C 100Khz MAX	100KHZ RMS CURRENT (MA)		
										25°C	85°C	125°C
50 VOLT RATING@+85C(33 VOLT RATING@+125C)												
TC*IX475DT	D	4.7	50	85	33	125	2.4	6	1.4	327	295	131
TC*IX475WT	W	4.7	50	85	33	125	2.4	6	1	316	285	126
TC*IX475VT	V	4.7	50	85	33	125	2.4	6	1.2	323	290	129
TC*IX685CT	C	6.8	50	85	33	125	3.4	6	1	332	298	133
TC*IX685DT	D	6.8	50	85	33	125	3.4	6	1	387	349	155
TC*IX685VT	V	6.8	50	85	33	125	3.4	6	0.9	373	335	149
TC*IX106DT	D	10	50	85	33	125	5	6	0.8	433	390	173
TC*IX106ET	E	10	50	85	33	125	5	6	0.8	454	409	182
TC*IX106HT	H	10	50	85	33	125	5	6	0.65	620	558	248
TC*IX156DT	D	15	50	85	33	125	7.5	6	0.6	500	450	200
TC*IX156ET	E	15	50	85	33	125	7.5	6	0.6	524	472	210
TC*IX156HT	H	15	50	85	33	125	7.5	6	0.6	645	581	258
TC*IX226ET	E	22	50	85	33	125	11	10	0.6	524	472	210
TC*IX226HT	H	22	50	85	33	125	11	8	0.6	645	581	258

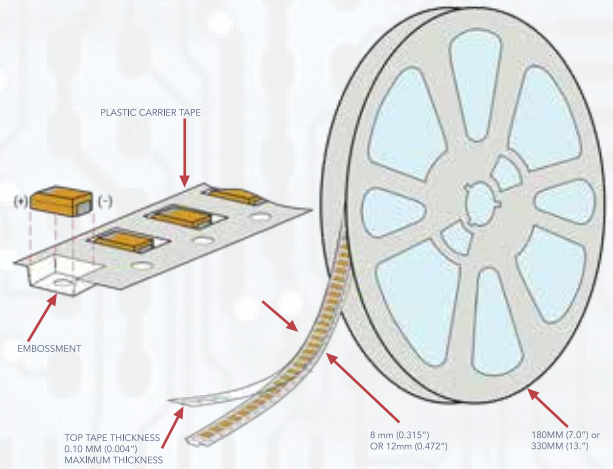
* Insert K for +/-10% tolerance and M for +/-20%. I Capacitance and DF are measured at 120Hz, 0.5V RMS with maximum DC bias of 2.0 volts.
DCL is measured at a rated voltage after 5 minutes. I Cal-chip reserves the right to substitute a higher voltage rated part
Additional values may be available, please contact your sales representative for more information.

TAPE AND REEL PACKAGING INFORMATION

Cal-Chip's molded tantalum and aluminum chip capacitor families are packaged in 8 and 12 mm plastic tape on 7" reels in accordance with EIA Standard 481: Embossed Carrier Taping of Surface Mount Components for Automatic Handling. This packaging system is compatible with all tape-fed automatic pick-and place systems.

CASE CODE		TAPE WIDTH (MM)	7" REEL
CCE	EIA		
A	3216-18	8	2,000
B	3528-21	8	2,000
C	6032-28	12	500
D	7343-31	12	500
E	7343-43	12	400/500
G	7361-43	12	400
H	7361-38	12	400

CASE CODE		TAPE WIDTH (MM)	7" REEL
CCE	EIA		
R	2012-12	8	2,000
P	2012-15	8	2,000
K	3216-10	8	3000
S	3216-12	8	2500
T	3528-12	8	2500
M	3528-15	8	2500
U	6032-15	12	1000
F	6032-20	12	1000
W	7343-15	12	1000
V	7343-20	12	1000



MS LEVEL 1

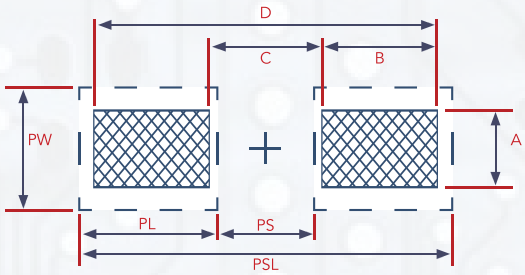
Cal-Chip Electronics Inc.
59 Steamwhistle Drive, Ivyland, PA 18974 | p. 215.942.8900 | www.calchip.com | quotes@calchip.com

QUALIFICATION TABLE

The component footprint is defined as the maximum board area taken up by the terminators.

The footprint dimensions are given by A, B, C and D in the diagram, which corresponds to W1 max., A max., S min. and L max. for the component (Please see data sheet). The footprint is symmetric about the center lines.

For reflow soldering, dimensions PL (Pad Length), PW (Pad Width), and PSL (Pad Set Length) have been calculated. For wave soldering the pad width (PWW) is reduced to less than the termination width to minimize the amount of solder pick up while ensuring that a good joint can be produced.



CASE CODE / SERIES		PSL	PL	PS	PW	PW_W
SMD, TC & TC-LSR SERIES	A	4.00 (0.157)	1.40 (0.055)	1.20 (0.047)	1.80 (0.071)	0.90 (0.035)
	B	4.00 (0.157)	1.40 (0.055)	1.20 (0.047)	2.80 (0.110)	1.60 (0.063)
	C	6.50 (0.256)	2.00 (0.079)	2.50 (0.098)	2.80 (0.110)	1.60 (0.063)
	D	8.00 (0.315)	2.00 (0.079)	4.00 (0.157)	3.00 (0.118)	1.70 (0.067)
	E	8.00 (0.315)	2.00 (0.079)	4.00 (0.157)	3.00 (0.118)	1.70 (0.067)
	F	6.50 (0.256)	2.00 (0.079)	2.50 (0.098)	2.80 (0.110)	1.60 (0.063)
	M	4.00 (0.157)	1.40 (0.055)	1.20 (0.047)	2.80 (0.110)	1.60 (0.063)
	K	4.00 (0.157)	1.40 (0.055)	1.20 (0.047)	1.80 (0.071)	0.90 (0.035)
	L	4.00 (0.157)	1.40 (0.055)	1.20 (0.047)	2.80 (0.110)	1.60 (0.063)
	N	2.70 (0.106)	0.95 (0.037)	0.80 (0.031)	1.60 (0.063)	0.80 (0.031)
	P	2.70 (0.106)	0.95 (0.037)	0.80 (0.031)	1.60 (0.063)	0.80 (0.031)
	R	2.70 (0.106)	0.95 (0.037)	0.80 (0.031)	1.60 (0.063)	0.80 (0.031)
	S	4.00 (0.157)	1.40 (0.055)	1.20 (0.047)	1.80 (0.071)	0.90 (0.035)
	T	4.00 (0.157)	1.40 (0.055)	1.20 (0.047)	2.80 (0.110)	1.60 (0.063)
	G	8.00 (0.315)	2.00 (0.079)	4.00 (0.157)	3.70 (0.145)	1.80 (0.071)
	V	8.00 (0.315)	2.00 (0.079)	4.00 (0.157)	3.70 (0.145)	1.80 (0.071)
	U	6.50 (0.256)	2.00 (0.079)	2.50 (0.098)	2.80 (0.110)	1.60 (0.063)
	X	8.00 (0.315)	2.00 (0.079)	4.00 (0.157)	3.00 (0.118)	1.70 (0.067)
	V	8.00 (0.315)	2.00 (0.079)	4.00 (0.157)	3.00 (0.118)	1.70 (0.067)
	5	8.00 (0.315)	2.00 (0.079)	4.00 (0.157)	3.00 (0.118)	1.70 (0.067)

QUALIFICATION TABLE

TEST	TC SERIES (TEMPERATURE RANGE -55oC to +125oC)										
	CONDITION			CHARACTERISTICS							
ENDURANCE	Apply rated voltage (Ur) at 85oC and / or category voltage (Uc) at 125oC for 2000 hours through a circuit impedance of ≤0.1Ω/V. Stabilize at room temperature for 1-2 hours before measuring.			Visual Examination	no visible damage						
				DCL	1.25 x initial limit						
				ΔC/C	within ±10% of initial value						
				DF	initial limit						
HUMIDITY	Store at 65oC and 95% relative humidity for 500 hours, with no applied voltage. Stabilize at room temperature and humidity for 1-2 hours before measuring.			Visual Examination	no visible damage						
				DCL	1.5 x initial limit						
				ΔC/C	within ±10% of initial value						
				DF	1.2 x initial limit						
TEMPERATURE STABILITY	STEP	TEMP. °C	DURATION (MIN)		+20°C	-55°C	+20°C	+85°C	+125°C	+20°C	
	1	+20	15	DCL	IL*	N/A	IL*	10 X IL*	12.5 X IL*	IL*	
	2	-55	15								
	3	+20	15	ΔC/C	N/A	+0/-10%	±5%	+10/-0%	+12/-0%	±5%	
	4	+85	15								
	5	+125	15	DF	IL*	1.5xIL*	IL*	1.5xIL*	2xIL*	IL*	
	6	+20	15								
SURGE VOLTAGE	Apply 1.3x category voltage (Uc) at 125oC for 1000 cycles of duration 6 min (30 sec charge, 5 min 30 sec discharge) through a charge / discharge resistance of 1000Ω			Visual Examination	no visible damage						
				DCL	initial limit						
				ΔC/C	within ±5% of initial value						
				DF	initial limit						
MECHANICAL SHOCK	MIL-STD-202, Method 213, Condition C			Visual Examination	no visible damage						
				DCL	initial limit						
				ΔC/C	within ±5% of initial value						
				DF	initial limit						
				ESR	initial limit						
VIBRATION	MIL-STD-202, Method 204, Condition D			Visual Examination	no visible damage						
				DCL	initial limit						
				ΔC/C	within ±5% of initial value						
				DF	initial limit						
				ESR	initial limit						